Introduction to Theoretical Computer Science

CS2490 - Session 8: From NFA to DFA (cont'd)

Pierre Cagne

Fall 2026

September 2, 2026

From NFA to DFA

How to transform any NFA into an *equivalent* DFA?

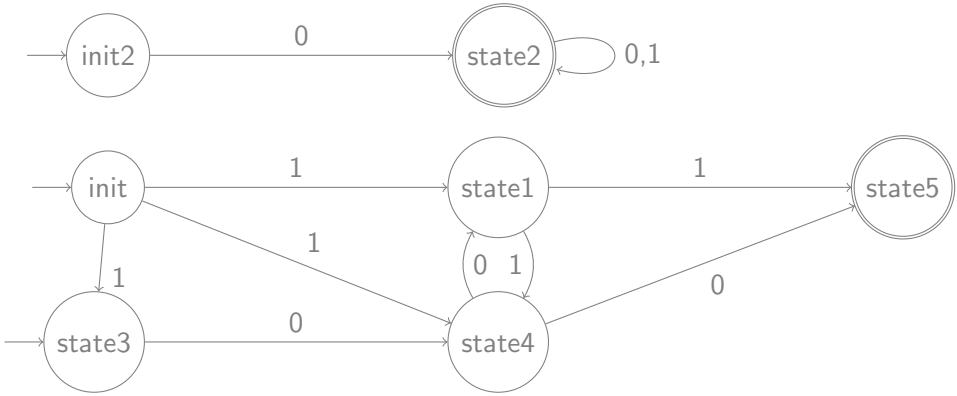
From NFA to DFA

How to transform any NFA into an *equivalent* DFA?

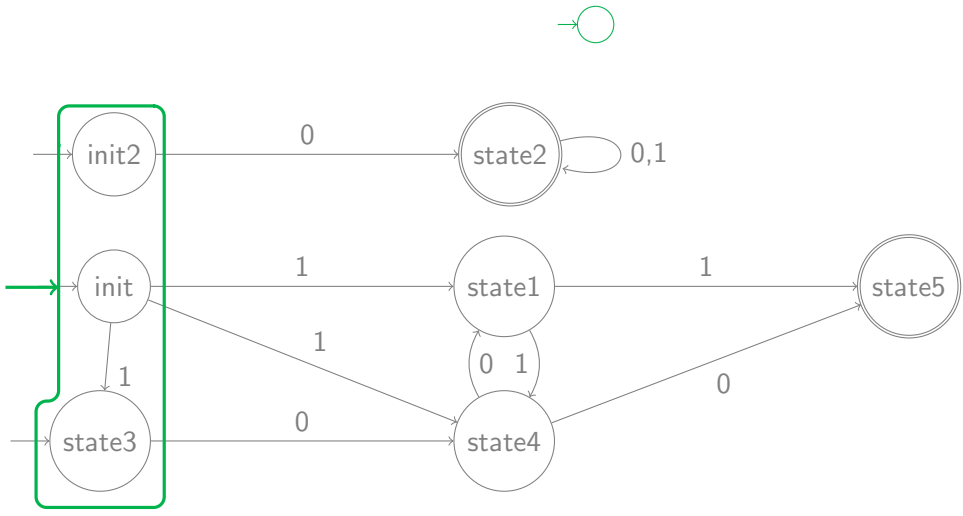
Given $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$, how to turn it into $\mathcal{A}^D = (\Sigma, Q^D, \delta^D, q_0^D, F^D)$ such that

$$\mathsf{L}(\mathcal{A}) = \mathsf{L}(\mathcal{A}^D) \quad ?$$

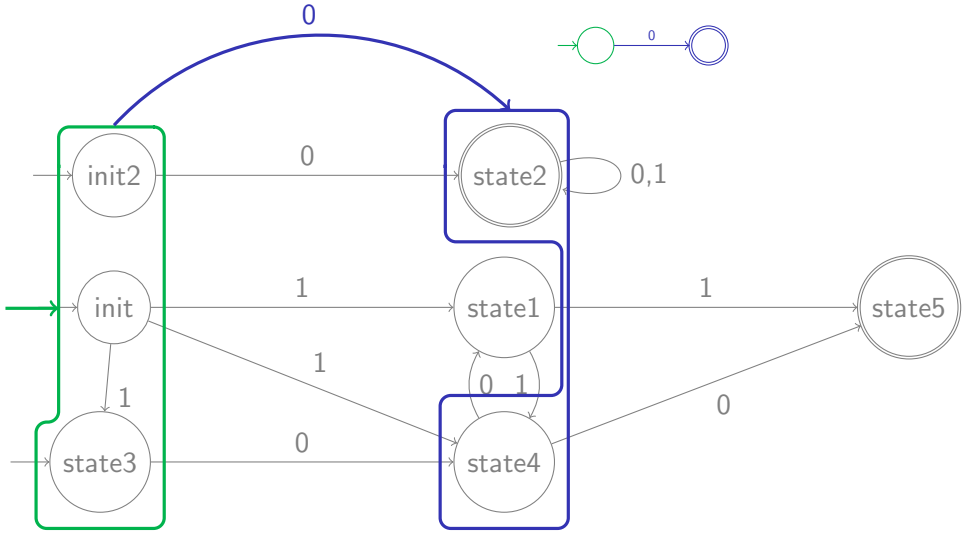
Trial by fire



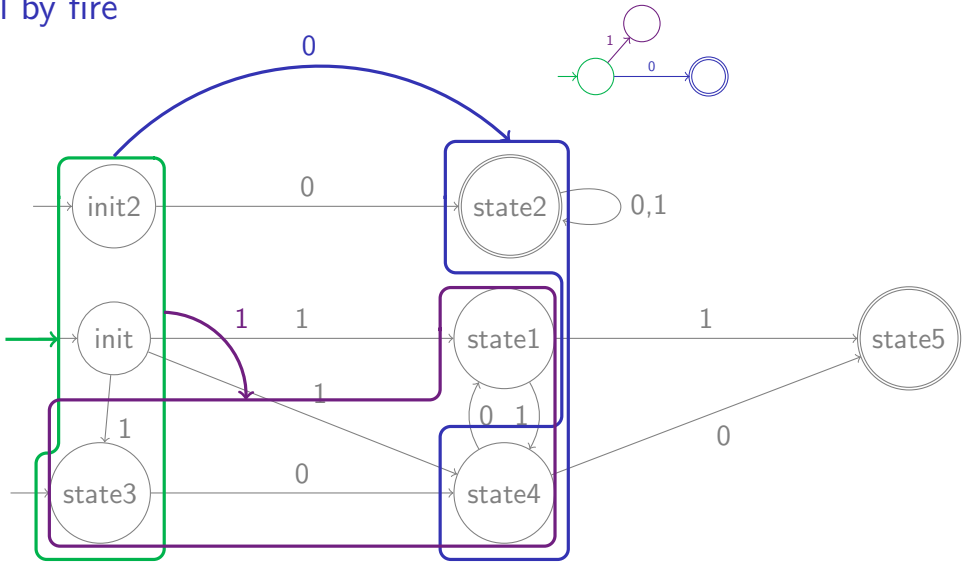
Trial by fire



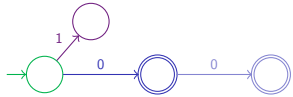
Trial by fire

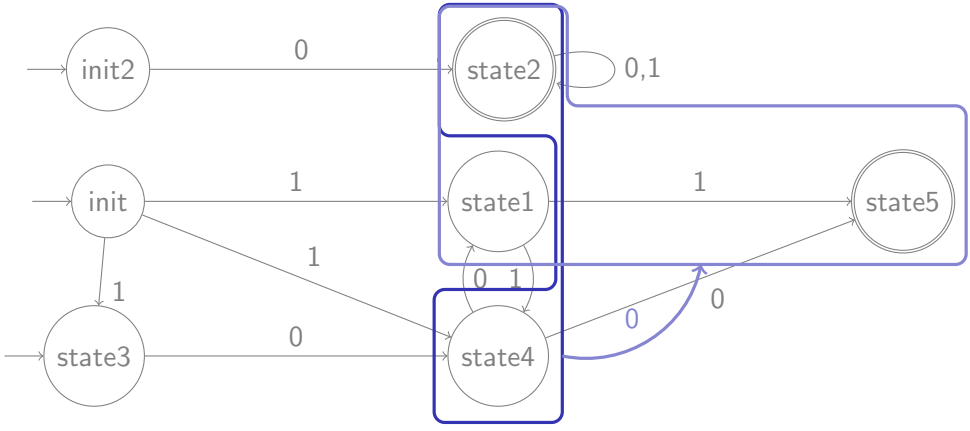


Trial by fire

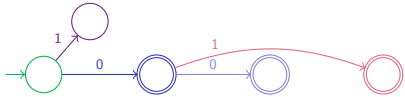


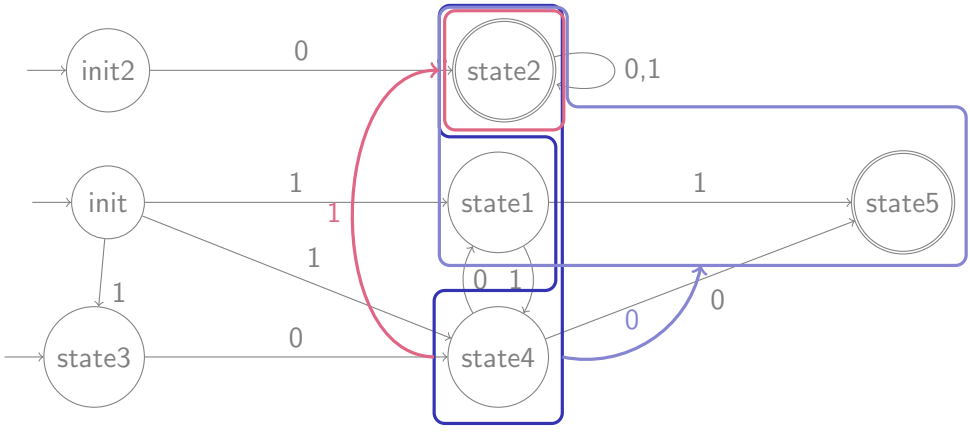
Trial by fire



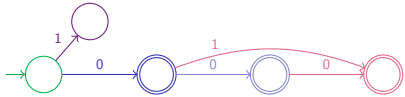


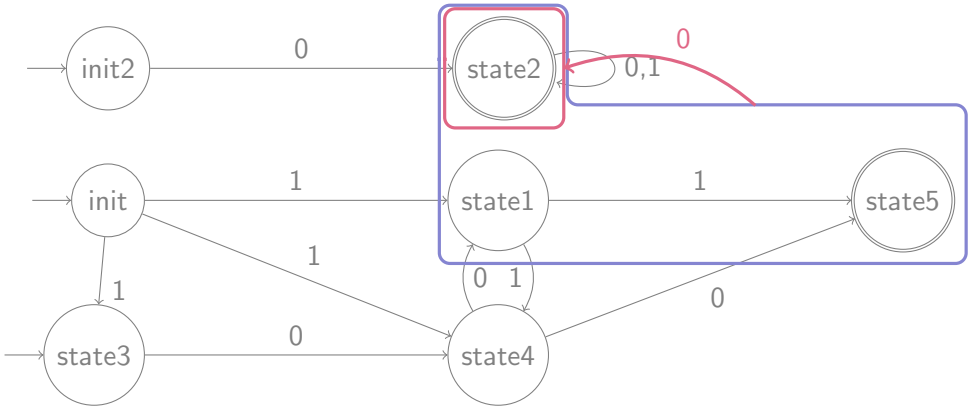
Trial by fire



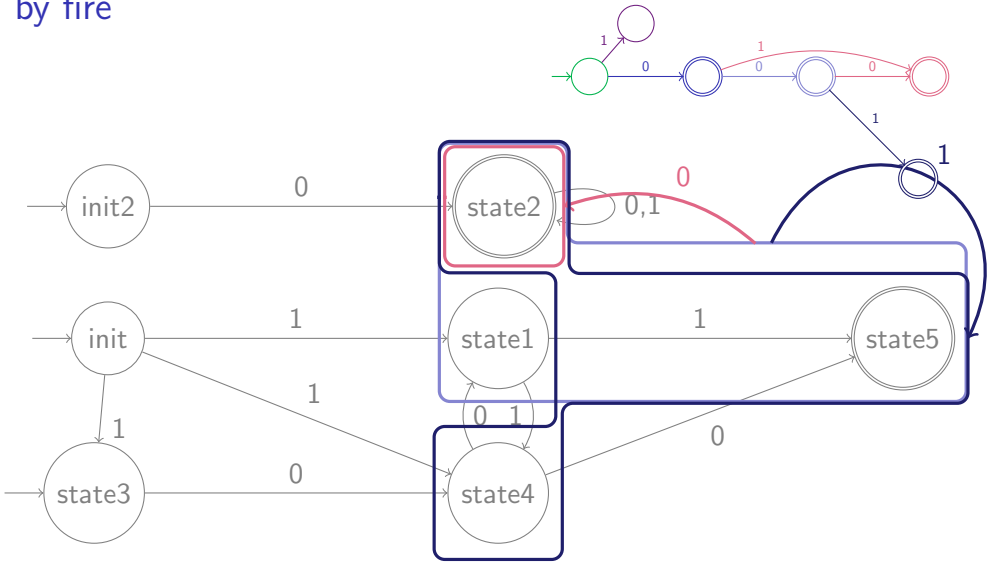


Trial by fire

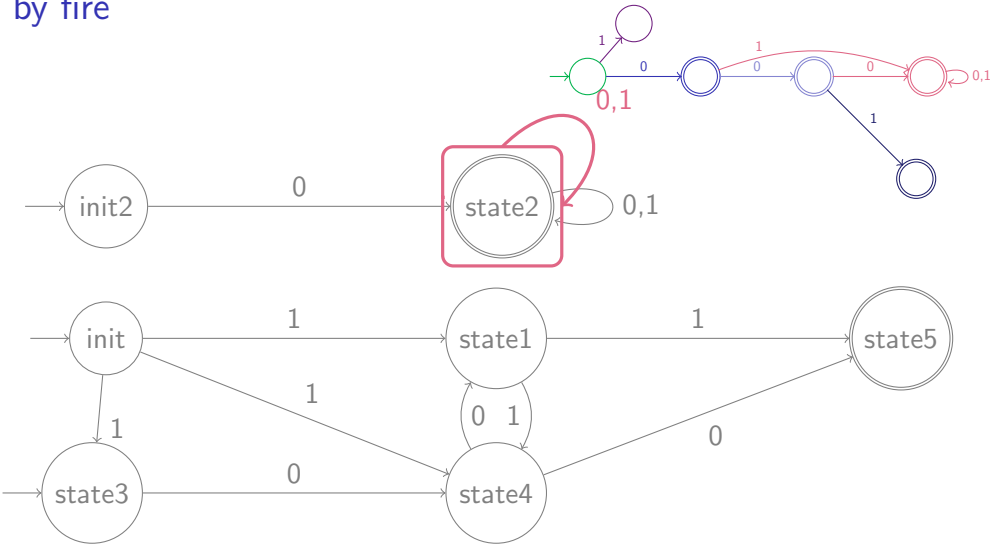




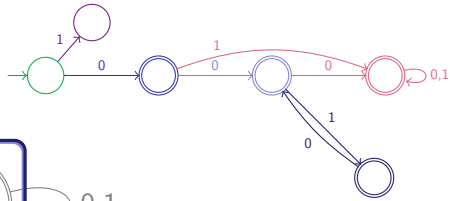
Trial by fire

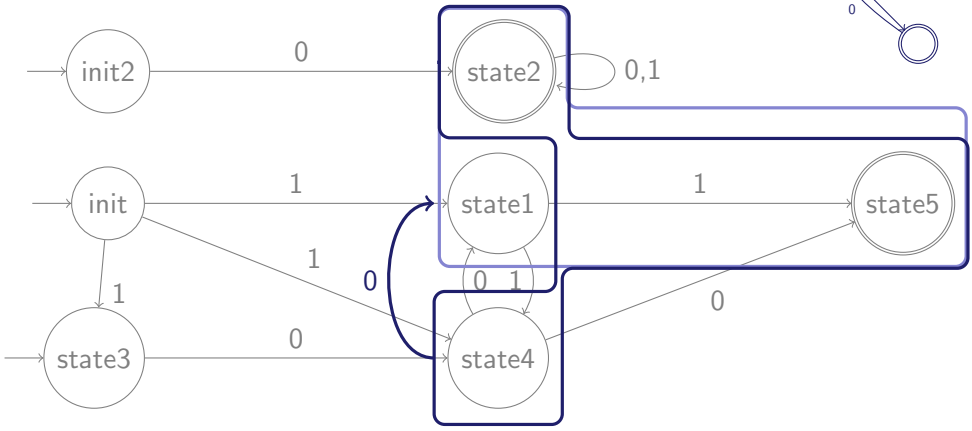


Trial by fire

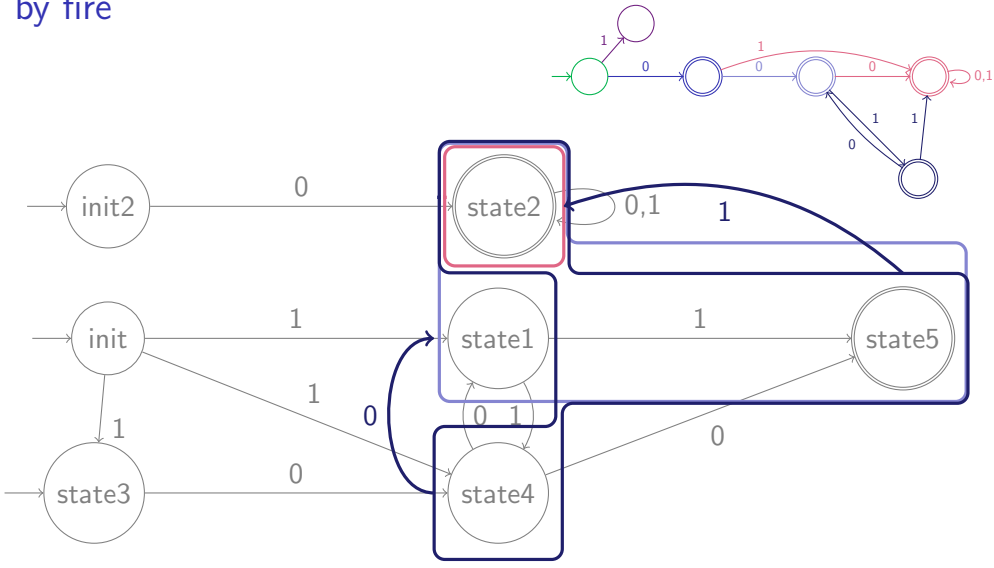


Trial by fire

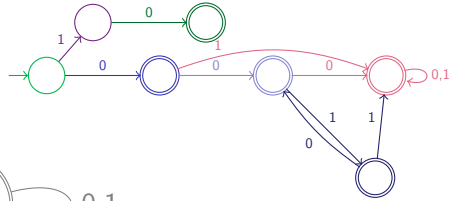


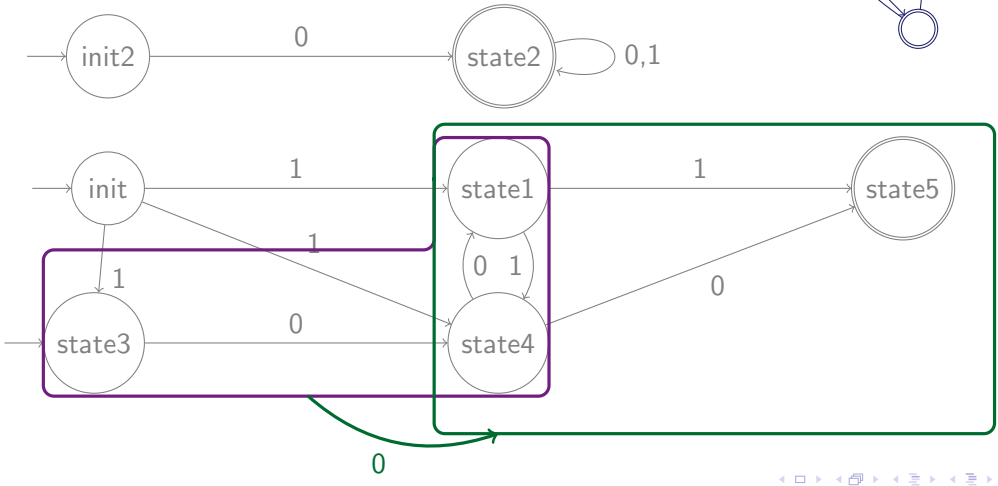


Trial by fire

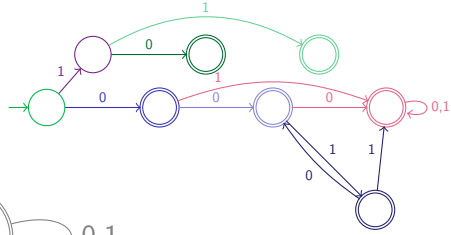


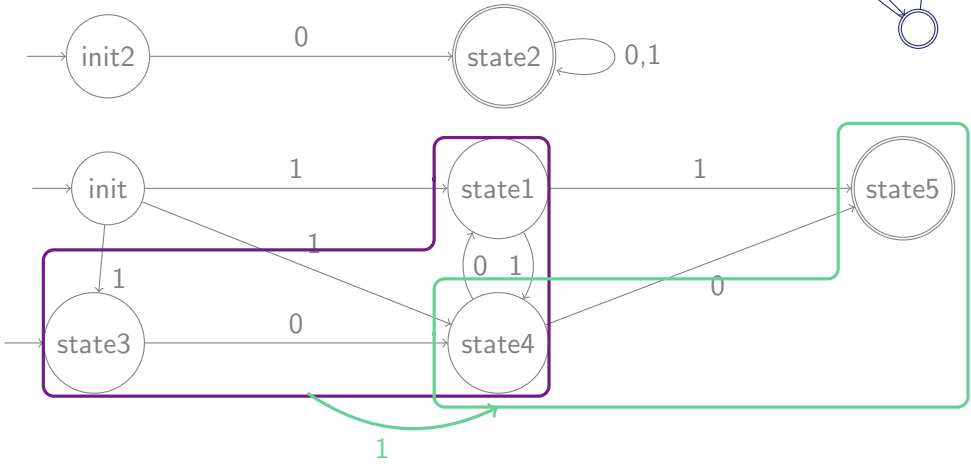
Trial by fire



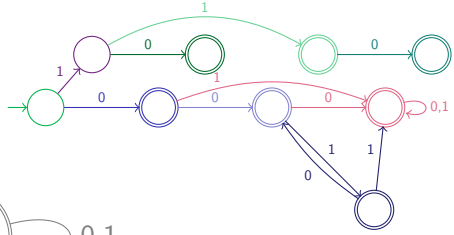


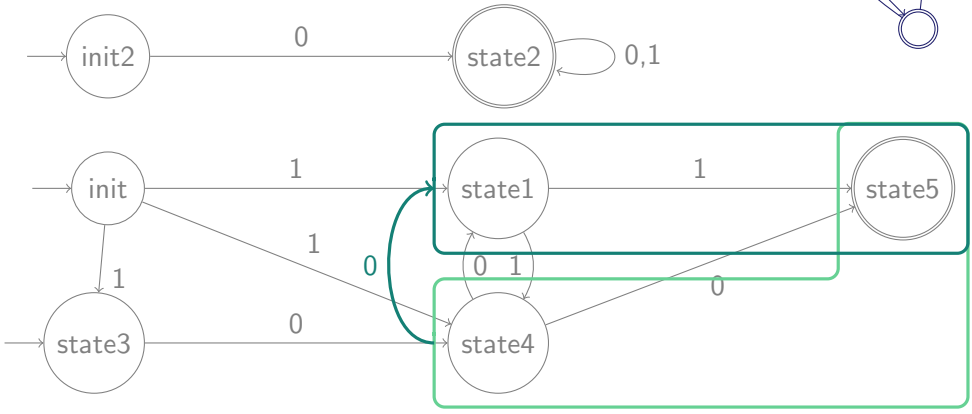
Trial by fire



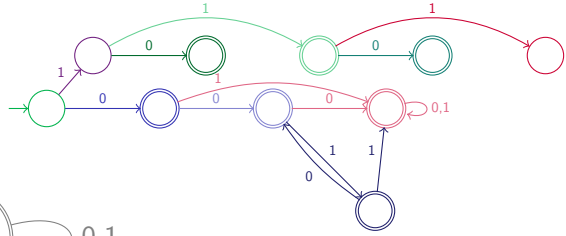


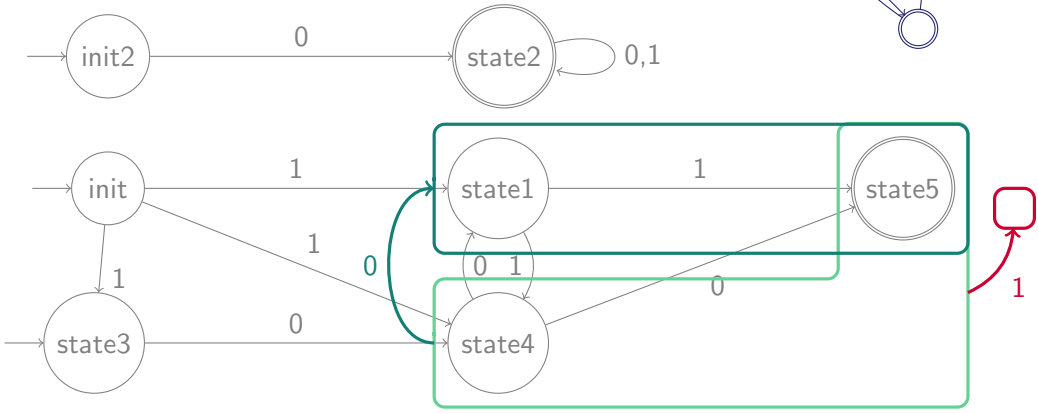
Trial by fire



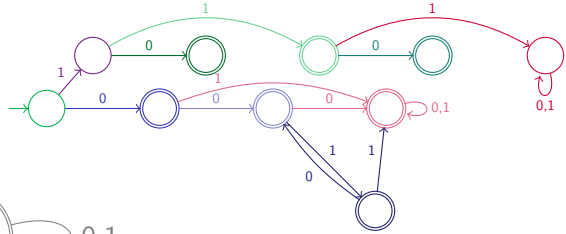


Trial by fire

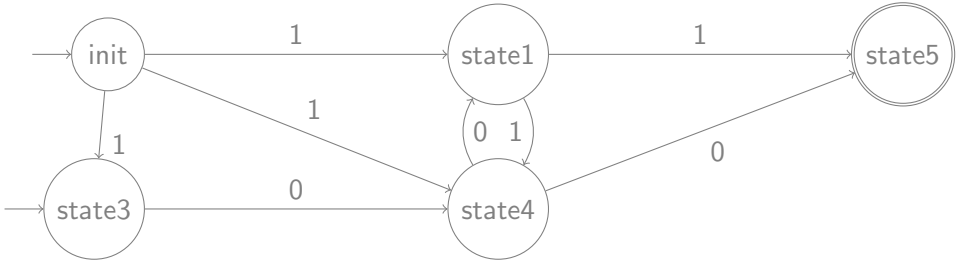




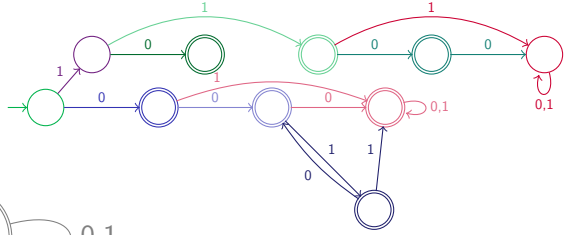
Trial by fire

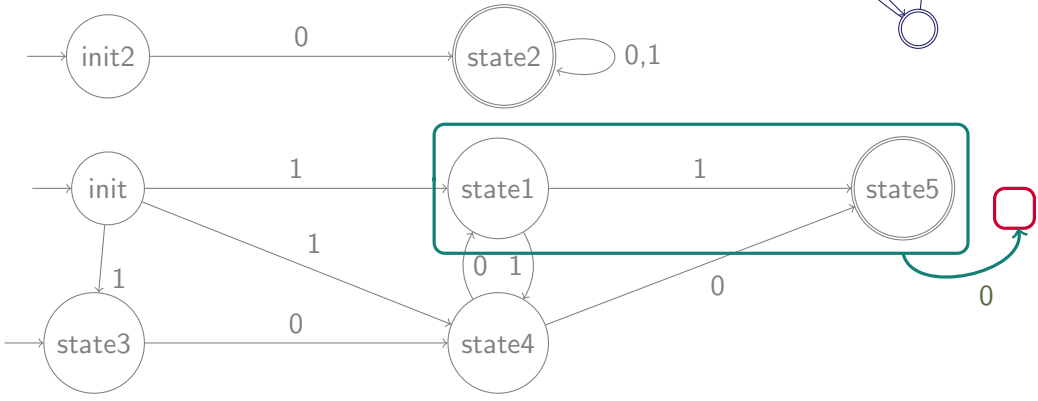




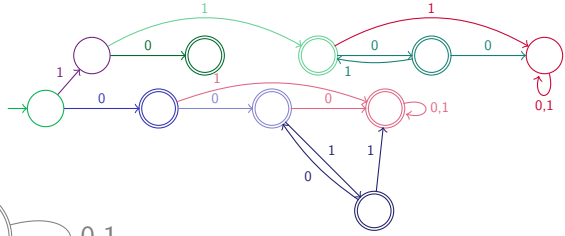


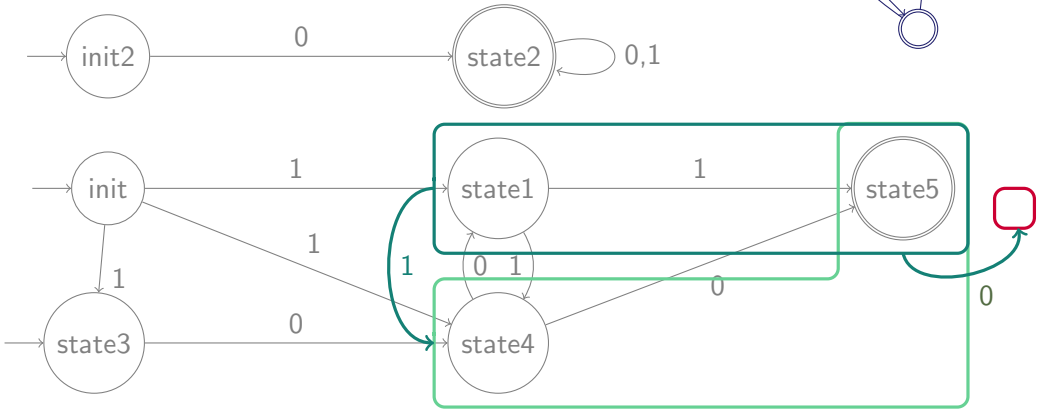
Trial by fire



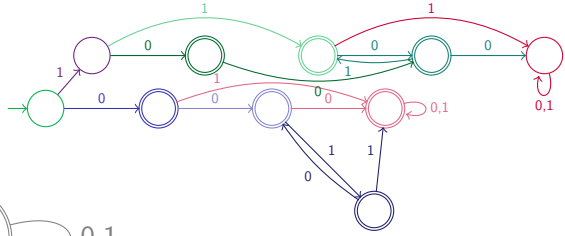


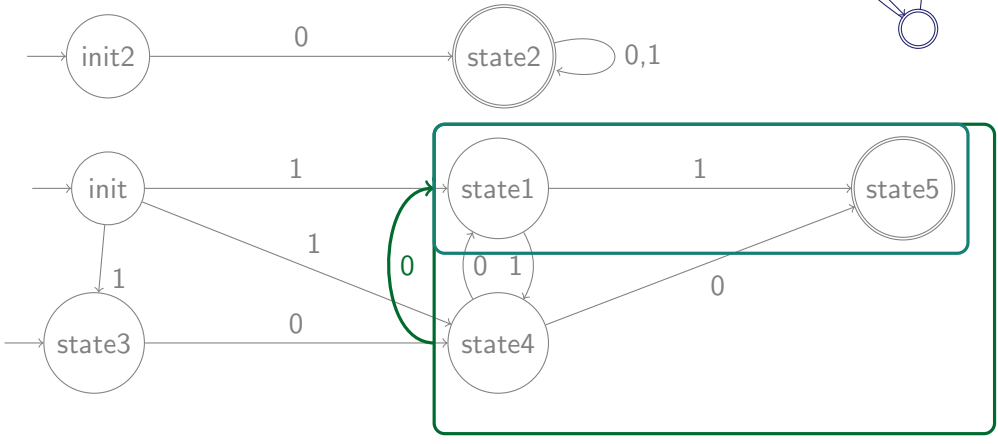
Trial by fire



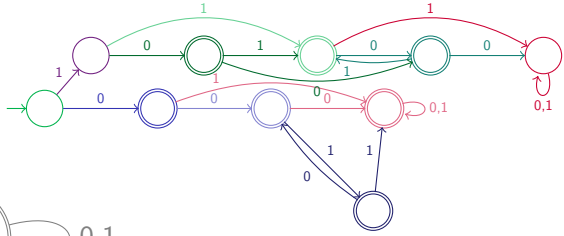


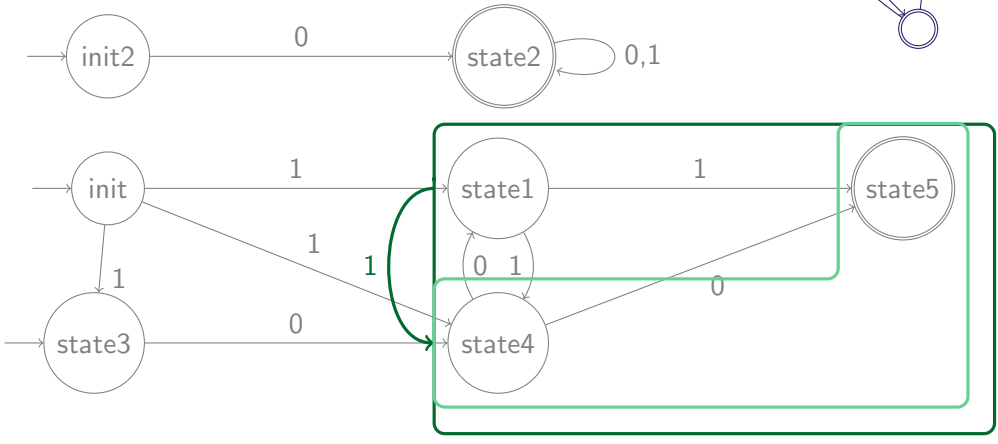
Trial by fire



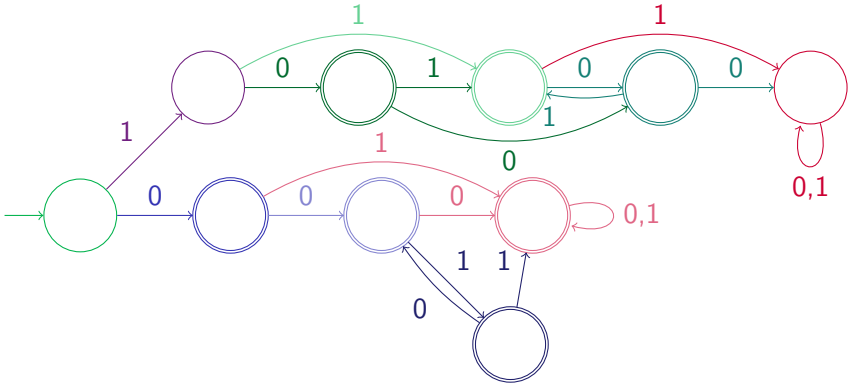


Trial by fire

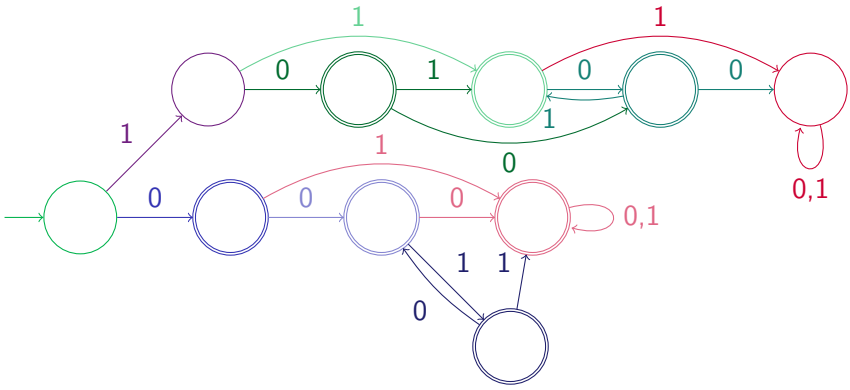




In the end

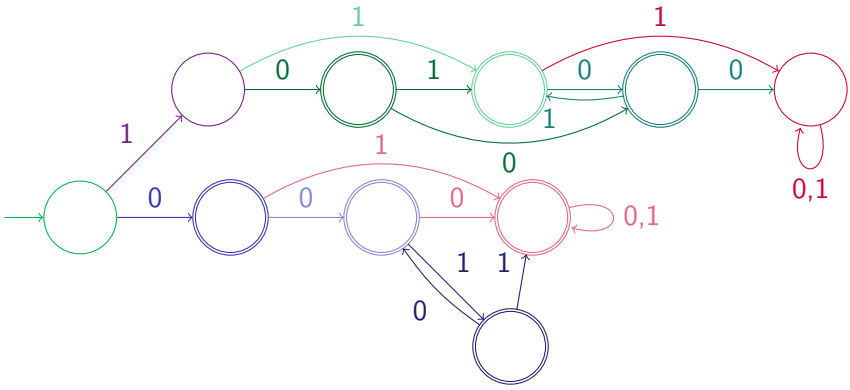


In the end



What are meaningful names for these states?

In the end



What are meaningful names for these states? Use the set of states each box contains!

Determinization algorithm

Let $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ be an NFA.

Determinization algorithm

Let $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ be an NFA.

Construct the DFA $\mathcal{A}^D = (\Sigma, Q^D, \delta^D, q_0^D, F^D)$ as follows.

Determinization algorithm

Let $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ be an NFA.

Construct the DFA $\mathcal{A}^D = (\Sigma, Q^D, \delta^D, q_0^D, F^D)$ as follows.

Q^D will be a subset of $\mathcal{P}(Q)$ (the powerset of Q), and initially contains only I.

Determinization algorithm

Let $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ be an NFA.

Construct the DFA $\mathcal{A}^D = (\Sigma, Q^D, \delta^D, q_0^D, F^D)$ as follows.

Q^D will be a subset of $\mathcal{P}(Q)$ (the powerset of Q), and initially contains only I.

While there exists $S \in Q^D$ and $a \in \Sigma$ such that $\delta^D(a, S)$ is undefined, do:

Determinization algorithm

Let $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ be an NFA.

Construct the DFA $\mathcal{A}^D = (\Sigma, Q^D, \delta^D, q_0^D, F^D)$ as follows.

Q^D will be a subset of $\mathcal{P}(Q)$ (the powerset of Q), and initially contains only I.

While there exists $S \in Q^D$ and $a \in \Sigma$ such that $\delta^D(a, S)$ is undefined, do:

- compute $\delta^D(a, S) = \bigcup_{q \in S} \Delta(a, q)$

Determinization algorithm

Let $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ be an NFA.

Construct the DFA $\mathcal{A}^D = (\Sigma, Q^D, \delta^D, q_0^D, F^D)$ as follows.

Q^D will be a subset of $\mathcal{P}(Q)$ (the powerset of Q), and initially contains only I.

While there exists $S \in Q^D$ and $a \in \Sigma$ such that $\delta^D(a, S)$ is undefined, do:

- compute $\delta^D(a, S) = \bigcup_{q \in S} \Delta(a, q)$
- add $\delta^D(a, S)$ to Q^D if not already present

Determinization algorithm

Let $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ be an NFA.

Construct the DFA $\mathcal{A}^D = (\Sigma, Q^D, \delta^D, q_0^D, F^D)$ as follows.

Q^D will be a subset of $\mathcal{P}(Q)$ (the powerset of Q), and initially contains only I.

While there exists $S \in Q^D$ and $a \in \Sigma$ such that $\delta^D(a, S)$ is undefined, do:

- compute $\delta^D(a, S) = \bigcup_{q \in S} \Delta(a, q)$
- add $\delta^D(a, S)$ to Q^D if not already present

Then define $q_0^D = I$ and $F^D = \{S \in Q^D \mid S \cap F \neq \emptyset\}$.

Equivalence

How to be sure that $\mathsf{L}(\mathcal{A}) = \mathsf{L}(\mathcal{A}^D)$?

Equivalence

How to be sure that $\mathsf{L}(\mathcal{A}) = \mathsf{L}(\mathcal{A}^D)$?

Let $w \in \Sigma^*$ be any word.

Equivalence

How to be sure that $\mathsf{L}(\mathcal{A}) = \mathsf{L}(\mathcal{A}^D)$?

Let $w \in \Sigma^*$ be any word.

Claim 1: For any computation of $\mathcal{A}$ on w ending in state q, the computation of $\mathcal{A}^D$ on w ends in a state S such that $q \in S$.

Equivalence

How to be sure that $\mathsf{L}(\mathcal{A}) = \mathsf{L}(\mathcal{A}^D)$?

Let $w \in \Sigma^*$ be any word.

Claim 1: For any computation of $\mathcal{A}$ on w ending in state q, the computation of $\mathcal{A}^D$ on w ends in a state S such that $q \in S$.

Proof: by induction on $|w|$.

Equivalence

How to be sure that $\mathsf{L}(\mathcal{A}) = \mathsf{L}(\mathcal{A}^D)$?

Let $w \in \Sigma^*$ be any word.

Claim 1: For any computation of $\mathcal{A}$ on w ending in state q, the computation of $\mathcal{A}^D$ on w ends in a state S such that $q \in S$.

Proof: by induction on $|w|$.

$\Longrightarrow$ If $w \in \mathsf{L}(\mathcal{A})$, then $w \in \mathsf{L}(\mathcal{A}^D)$.

Equivalence

How to be sure that $\mathsf{L}(\mathcal{A}) = \mathsf{L}(\mathcal{A}^D)$?

Let $w \in \Sigma^*$ be any word.

Claim 1: For any computation of $\mathcal{A}$ on w ending in state q, the computation of $\mathcal{A}^D$ on w ends in a state S such that $q \in S$.

Proof: by induction on $|w|$.

$\Longrightarrow$ If $w \in \mathsf{L}(\mathcal{A})$, then $w \in \mathsf{L}(\mathcal{A}^D)$.

Claim 2: If the computation of $\mathcal{A}^D$ on w ends in a state S, then for any $q \in S$ there exists a computation of $\mathcal{A}$ on w ending in q.

Equivalence

How to be sure that $\mathsf{L}(\mathcal{A}) = \mathsf{L}(\mathcal{A}^D)$?

Let $w \in \Sigma^*$ be any word.

Claim 1: For any computation of $\mathcal{A}$ on w ending in state q, the computation of $\mathcal{A}^D$ on w ends in a state S such that $q \in S$.

Proof: by induction on $|w|$.

$\Longrightarrow$ If $w \in \mathsf{L}(\mathcal{A})$, then $w \in \mathsf{L}(\mathcal{A}^D)$.

Claim 2: If the computation of $\mathcal{A}^D$ on w ends in a state S, then for any $q \in S$ there exists a computation of $\mathcal{A}$ on w ending in q.

Proof: by induction on $|w|$.

Equivalence

How to be sure that $\mathsf{L}(\mathcal{A}) = \mathsf{L}(\mathcal{A}^D)$?

Let $w \in \Sigma^*$ be any word.

Claim 1: For any computation of $\mathcal{A}$ on w ending in state q, the computation of $\mathcal{A}^D$ on w ends in a state S such that $q \in S$.

Proof: by induction on $|w|$.

$\Longrightarrow$ If $w \in \mathsf{L}(\mathcal{A})$, then $w \in \mathsf{L}(\mathcal{A}^D)$.

Claim 2: If the computation of $\mathcal{A}^D$ on w ends in a state S, then for any $q \in S$ there exists a computation of $\mathcal{A}$ on w ending in q.

Proof: by induction on $|w|$.

$\Longrightarrow$ If $w \in \mathsf{L}(\mathcal{A}^D)$, then $w \in \mathsf{L}(\mathcal{A})$.

Proof of Claim 1

Proof of Claim 1

- **Base case:** $w = \varepsilon$. The computation of $\mathcal{A}^D$ on ε ends in the initial state I. Any computation of $\mathcal{A}$ on ε ends in some initial state q, and so $q \in I$, as wanted.

Proof of Claim 1

- **Base case:** $w = \varepsilon$. The computation of $\mathcal{A}^D$ on ε ends in the initial state I. Any computation of $\mathcal{A}$ on ε ends in some initial state q, and so $q \in I$, as wanted.
- **Inductive step:** Assume the claim holds for all words of length n. Let $w' = wa$ with $a \in \Sigma$ and w of length n. Suppose a computation of $\mathcal{A}$ on w' ends in q'. This computation starts with a computation on w that ends in some state q, and the last step of the full computation on w' leads to a state $q' \in \Delta(a, q)$. By the induction hypothesis, the computation of $\mathcal{A}^D$ on w ends in a state S with $q \in S$. By the determinization algorithm, the computation of $\mathcal{A}^D$ on w' ends in $S' = \bigcup_{p \in S} \Delta(a, p)$ that contains $\Delta(a, q)$ as a subset and so $q' \in S'$. Thus, the claim holds for w'.

Proof of Claim 2

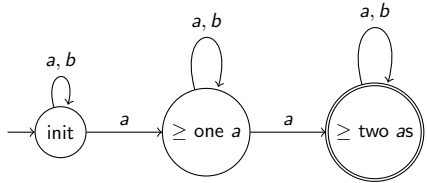
Proof of Claim 2

- **Base case:** $w = \varepsilon$. The computation of $\mathcal{A}^D$ on ε ends in the initial state I. Any state $q \in I$ is initial, and so the computation that only contains q is a computation of $\mathcal{A}$ on ε ending in q, as wanted.

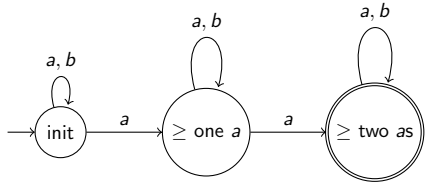
Proof of Claim 2

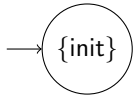
- **Base case:** $w = \varepsilon$. The computation of $\mathcal{A}^D$ on ε ends in the initial state I. Any state $q \in I$ is initial, and so the computation that only contains q is a computation of $\mathcal{A}$ on ε ending in q, as wanted.
- **Inductive step:** Assume the claim holds for w of length n. Let $w' = wa$ with $a \in \Sigma$ and w of length n. Suppose the computation of $\mathcal{A}^D$ on w' ends in state S'. By the determinization algorithm, $S' = \bigcup_{q \in S} \Delta(a, q)$, where S is the state reached after w. For any $q' \in S'$, there exists $q \in S$ such that $q' \in \Delta(a, q)$. By the induction hypothesis, there is a computation of $\mathcal{A}$ on w ending in q, and then a transition to q' reading a. Thus, there is a computation of $\mathcal{A}$ on w' ending in q'. The claim thus holds for w'.

Let's practice

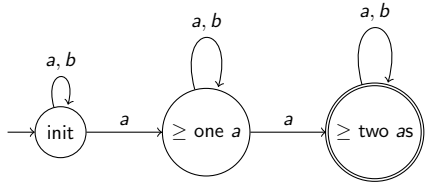


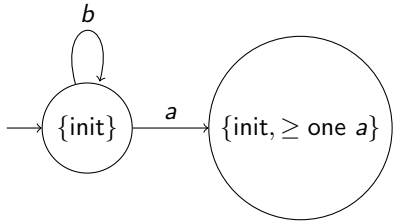
Let's practice



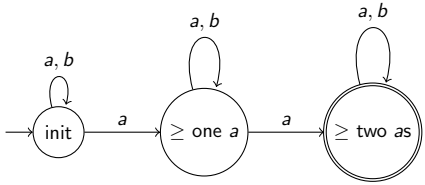


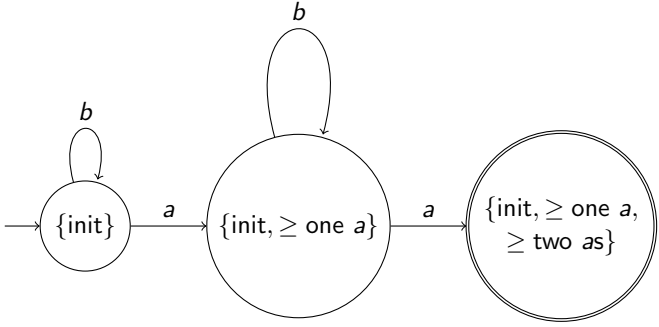
Let's practice



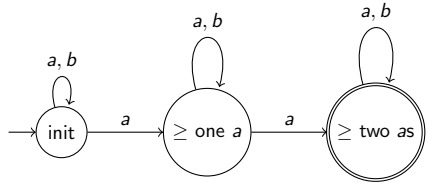


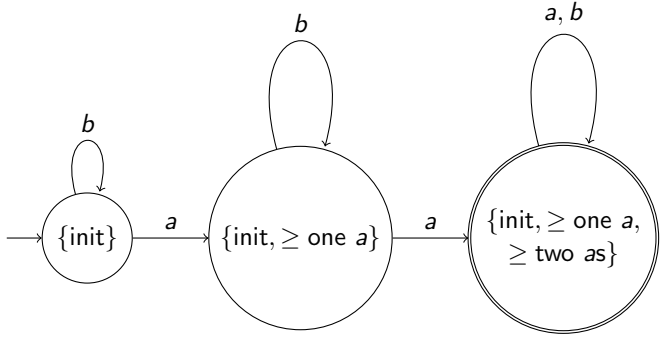
Let's practice



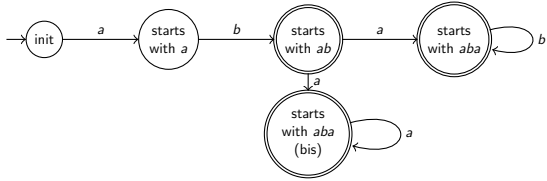


Let's practice

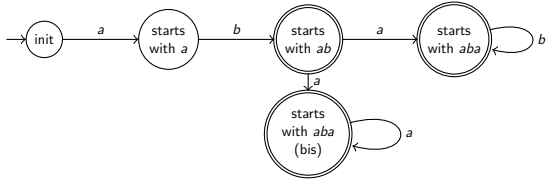


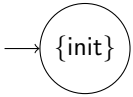


Let's practice

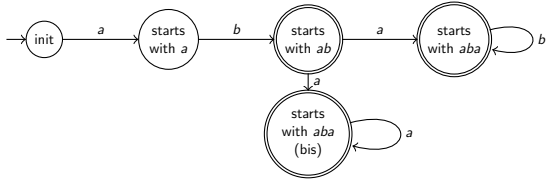


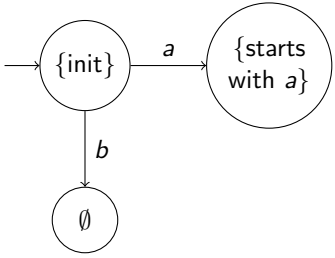
Let's practice



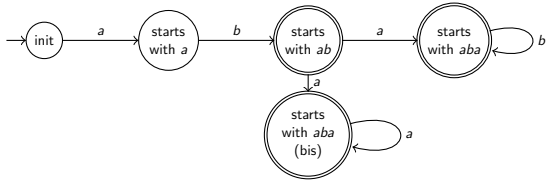


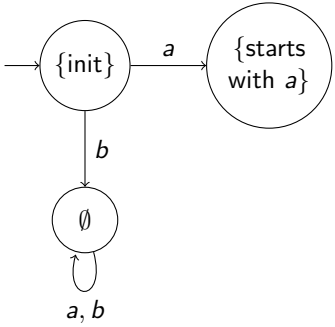
Let's practice



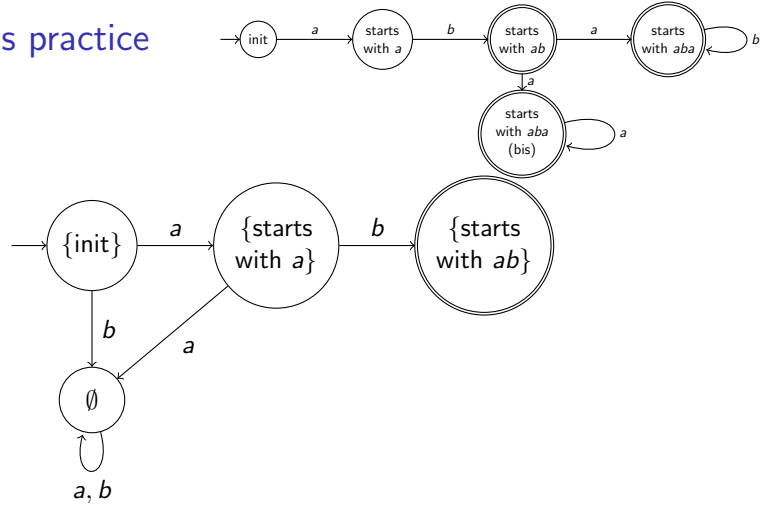


Let's practice

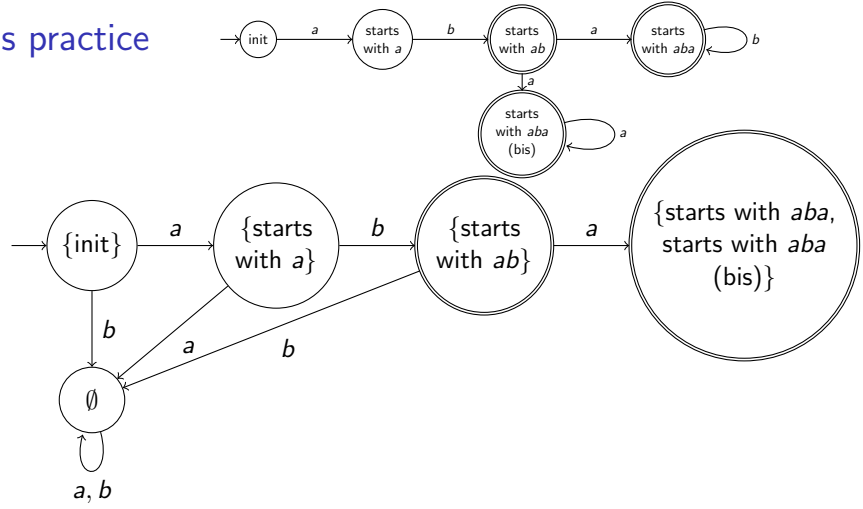




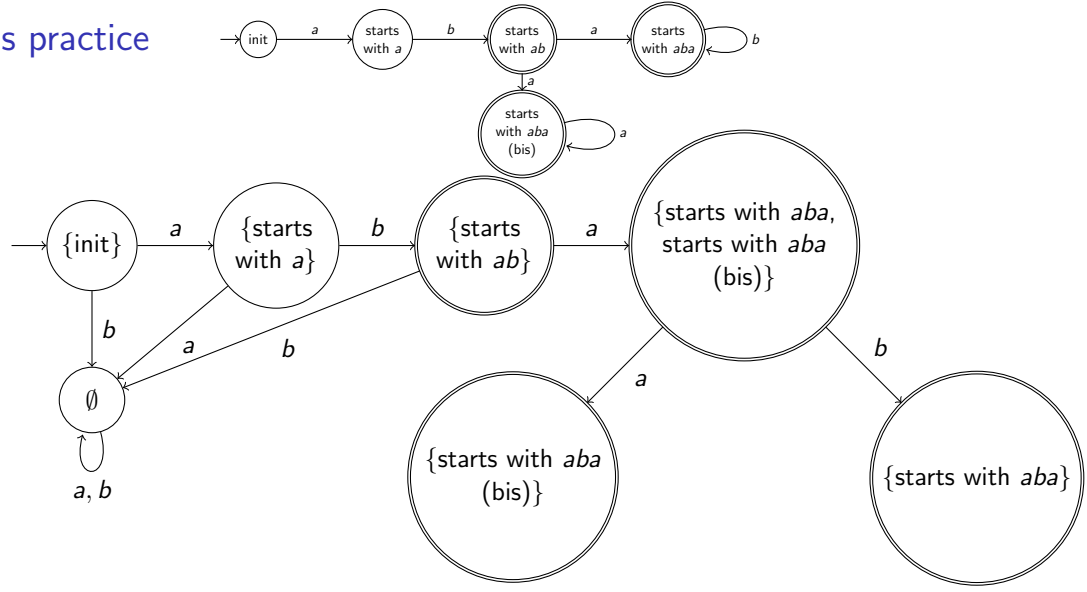
Let's practice



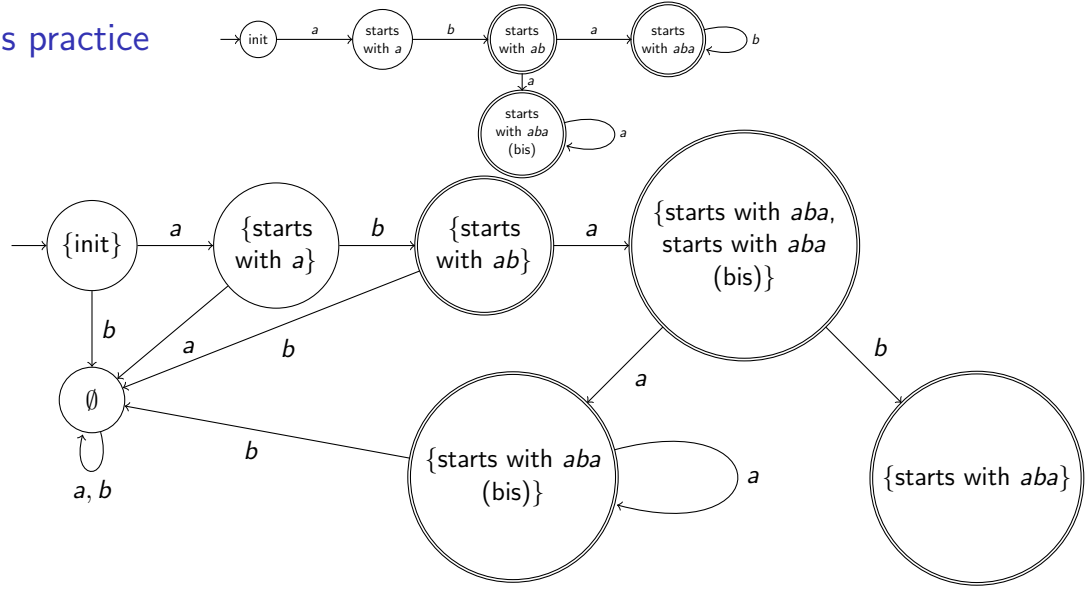
Let's practice



Let's practice



Let's practice



Let's practice

